College of Computer Science and Information Technology
Department of Computer Engineering

ARTI403: Image Processing

Lab 1: Basics of Programming with Python

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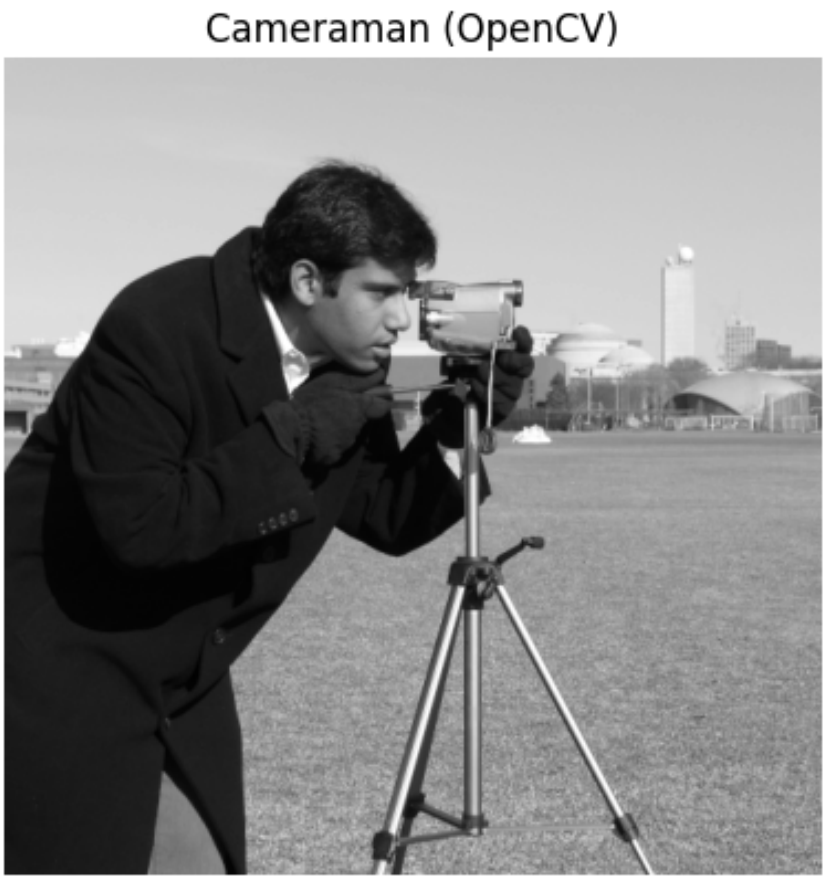
Import the libraries

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.cm as cm
from PIL import Image
```

Task 2: Loading and Visualizing Images

```
# load the image with OpenCV
img = cv2.imread('images/cameraman.tif', cv2.IMREAD_GRAYSCALE)

plt.imshow(img, cmap='gray')
plt.title('Cameraman (OpenCV)')
plt.axis('off')
plt.show()
```




```
# load the image with PIL
img2 = Image.open('images/lena_gray_256.tif')

plt.imshow(img2, cmap=cm.Greys_r)
plt.title('Lena (PIL)')
plt.axis('off')
plt.show()
```



Task 3: Image Storing

```
cv2.imwrite('new_image.jpg', img)
img2.save('new_image2.jpg')
```

Task 4: Display Image as an Array

```
# the OpenCV image is already a numpy array
print(img.shape)
print(img)
```

```
(512, 512)
[[200 200 200 ... 189 190 190]
 [200 199 199 ... 190 190 190]
 [199 199 199 ... 190 190 190]
 ...
 [ 25  25  27 ... 139 122 147]
 [ 25  25  26 ... 158 141 168]
 [ 25  25  27 ... 151 152 149]]
```

```
# convert the PIL image to a numpy array
img_array = np.array(img2)
print(img_array.shape)
print(img_array)
```

```
(256, 256)
[[199 199 200 ... 190 190 190]
 [199 199 200 ... 190 190 190]
 [200 200 199 ... 191 191 191]
 ...
 [ 25  25  24 ... 136 143 118]
 [ 26  25  26 ... 147 154 144]
 [ 25  25  25 ... 146 146 151]]
```

Assessment: NumPy operations on the image array

```
print('min:', np.min(img_array))
print('max:', np.max(img_array))
print('mean:', np.mean(img_array))
print('std:', np.std(img_array))
```

```
min: 0
max: 255
mean: 129.06893920898438
std: 73.04908779785245

# negative, brighter, crop and flip
negative = 255 - img_array
brighter = np.clip(img_array.astype(int) + 50, 0, 255)
cropped = img_array[50:200, 50:200]
flipped = np.fliplr(img_array)

fig, axes = plt.subplots(1, 4, figsize=(16, 4))
for ax, im, t in zip(axes, [negative, brighter, cropped, flipped],
                     ['Negative', 'Brighter (+50)', 'Cropped', 'Flipped']):
    ax.imshow(im, cmap='gray', vmin=0, vmax=255)
    ax.set_title(t)
    ax.axis('off')
plt.show()
```

